

23.2 A 2-Channel 800Gb/s Transceiver for Coherent-Lite Applications with <300ns Latency in 5nm FinFET

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Abstract

A full transceiver suitable for coherent-lite optical communications over 2-to-20km distances is presented. The low-latency transceiver combines the advantages of advanced coherent modulation with the benefits and manufacturing scale of O-band optics to deliver a power-

and cost-effective module. Fabricated in 5nm FinFET, it comprises two 400Gb/s/L transmitters and receivers to realize a 2-channel DP-16QAM 800Gb/s link tailored to the market of distributed campus data center interconnects.

The rapid growth of AI and cloud services is increasing data center complexity, leading to challenges in physical space and power delivery. Data-center architectures are thus shifting from large-scale facilities to distributed campus-based architectures, increasing the need for optimized interconnects in the 2-to-40km range.

Currently, intensity modulation direct detection (IMDD) is the predominant technology in data centers, while coherent interconnects are used for regional connectivity (Fig. 23.2.1). However, IMDD hits a performance limit as reach and link-speed requirements increase. Concurrently, traditional coherent solutions become extremely power-inefficient at shorter distances. The coherent-lite solution introduced in this paper combines the benefits of IMDD and coherent technologies [1,2], enabling low-latency and power-optimized 2-40km communications at 400Gb/s per wavelength with DP-16QAM modulation. By leveraging the favorable trade-off between chromatic dispersion and optical attenuation in the O-band to limit DSP equalization complexity, the presented architecture achieves over 10× lower latency than conventional coherent systems and 2× higher power efficiency, positioning it as a compelling candidate for next-generation AI interconnects. The transceiver also integrates a software-reconfigurable high-swing optical modulator driver, reducing module power consumption and component count. Two channels are combined in a single die, resulting in an 800Gb/s aggregate data rate, suitable for integration into 400G/800G/1.6T pluggable modules. The device is fabricated using a 5nm FinFET process.

Figure 23.2.1 shows the block diagram of a 400Gb/s coherent-lite channel. A single clocking section (CLK) drives four line transmitter (LTX) or line receiver (LRX) lanes in a quad configuration. Each quad supports a 400Gb/s data-stream at the electrical interface, handling horizontal and vertical polarizations (X, Y) and in-phase and quadrature-phase modulations (I, Q). DP-16QAM modulation and demodulation are performed in the optical domain, utilizing highly integrated silicon photonics O-band optics. In the LRX path, external transimpedance amplifiers (TIAs) provide low noise amplification and maximize receiver sensitivity.

The quad LTX block diagram (Fig. 23.2.2) includes a fractional-N PLL, generating a 31.25GHz half-rate clock, and per-lane phase interpolators (PI). The PI output clocks are distributed across the four lanes (XI, XQ, YI, YQ). To ensure the synchronization required for coherent modulation, a skew-alignment calibration system compensates on-die and off-die skews by operating on the PI. The TX DSP includes convolution interleaving and inner FEC encoding. XI, XQ, YI, YQ data streams are generated by the DP-16QAM mapper and pilot symbol insertion. A 7-tap FIR filter enables signal equalization and mitigation of reflection artifacts [3]. The block diagram of an LTX lane is shown in Fig. 23.2.2. The 7b × 64 data stream from the DSP is fed to the serializer, which performs the 64:1 parallel to serial conversion. A full-speed pre-driver delivers the signal to a 7b DAC-based driver to transmit data in PAM-4 format. To address clock distribution impairments, a duty cycle correction (DCC) mechanism is adopted. Power efficiency and optimized jitter performance across PVT are achieved through local automatic voltage scaling (AVS), dynamically adjusting the supply levels.

A Class-AB architecture [4,5], driving both the NMOS and PMOS sections of the driver and pushing and pulling current into a differential termination, has been chosen for the pseudo-differential DAC driver to improve power efficiency (Fig. 23.2.2). The combination of AC-coupler-based level shifters and differential latches, used inside each bit within the DAC, enables consistent performance at high frequency and immunity to wandering effects [6]. Double cascodes provide low-impedance current collection extending the bandwidth and shielding the DAC from high-voltage swings. Dynamic biasing ensures long-term reliability [7]. The DAC driver slices feed into a differential termination (RTERM) and TCOIL. The RTERM is reconfigurable to optimize performance with a variety of Mach-Zehnder modulator (MZM) designs.

The DAC driver is reconfigurable as either an integrated silicon photonics (SiPho) driver, delivering 4V_{ppd} to a 66Ω differential load, or a standard (STD) driver, delivering 1V_{ppd} to a 100Ω load. In the former case, the integrated circuit directly drives a SiPho MZM modulator, while in the latter an external high-swing driver needs to be added at the module level, reducing the overall efficiency. The driver can be reconfigured to STD mode by operating on PMOS-bias and PMOS-NMOS cascode blocks and reducing LTX voltage supply.

Figure 23.2.3 shows the LRX architecture which operates with an 8/7 oversampling ratio using a half rate 35.7GHz clock. The LRX PLL architecture is the same as the LTX PLL, except for a dedicated VCO design. The PLL output clock is distributed to the XI, XQ, YI, YQ lanes. A coarse skew detection and alignment scheme ensures clock and data synchronization across lanes [8]. Residual skews are then compensated in the digital equalization engine.

The LRX lane, illustrated in Fig. 23.2.3, has been designed to achieve an SNDR in excess of 32dB up to 30GHz. An external TIA converts current into voltage and drives the analog front end. The internal 100Ω termination is DC coupled directly to the input pads, while the VGA input is AC coupled. The VGA is implemented as a single-stage PMOS-based CML amplifier utilizing programmable MOS transistors for resistive degeneration [3]. Bandwidth extension is achieved through cross-coupled MOM capacitors, implementing a negative capacitance at the VGA input, and inductive peaking. The gain is programmable between -2.5dB and 6dB. The VGA drives a 71.4GS/s 7b ADC with a 64-way time-interleaved architecture based on a partial-loop-unrolled SAR structure [9]. It adopts a hierarchical scheme, where 8 track-and-hold units each feeds 8 sub-ADCs. A dedicated clock generation block (CLK GEN) produces the required 8Ts and 64Ts sampling phases from the PLL clock. This block also includes delay lines to fine tune the track-and-hold units, achieving timing alignment with sub-100fs precision.

The RX DSP (Fig. 23.2.3) implements a low-power architecture suitable for low-latency coherent-lite modules. A 4×4 real multiple-input-multiple-output (MIMO) equalizer handles bandwidth and fiber dispersion effects, as well as carrier and polarization rotations introduced by the coherent optical modulator/demodulator and the optical fiber. It is implemented as a poly-phase filter with 8/7 oversampled inputs and symbol-rate outputs. This architecture is suitable for O-band designs with low chromatic dispersion tolerance requirements and, compared to coherent DSPs [10], provides lower latency and better immunity to skew, gain and quadrature phase imbalances between the four input lanes, as well as lower power dissipation. A carrier frequency/phase recovery block uses known pilot symbols, spaced every 64 symbols, to handle LO frequency offset and laser linewidth effects. A timing recovery block, controlling the PLL, generates a local LRX sampling clock that is synchronous to the transmitter clock. A post-processing block improves robustness to optical impairments and is key to the design of low-cost and low-power coherent-lite optical modules. It features adaptive level tracking and skew adjustment filters to provide robustness to transmit gain, skew, offset and quadrature phase imbalance impairments. A high-performance, low-power and low-latency forward error-correction (FEC) decoder is used to recover the transmit bits. The FEC concatenates an inner BCH(126,110) code with an outer RS(544,514) code, which is used in IEEE 802.3 compliant chip-to-module (C2M) interfaces. The inner code is added without termination of the RS code in either the TX or the RX paths to minimize latency. The inner code is decoded using a simple soft-input/hard-output Chase decoder [11], which results in an input BER threshold of 1×10⁻², to achieve <2.2×10⁻⁴ BER at the C2M interface and <1×10⁻¹⁵ BER after RS FEC decoding in the host device. The latency from ADC outputs to the FEC decoder input is <20ns.

A photograph of the 5nm FinFET die is shown in Fig. 23.2.7. It features 8×100Gb/s PAM-4 TX and RX lanes on the host side and 2×400Gb/s coherent-lite channels on the line side.

Figure 23.2.4 presents the 125Gb/s PAM-4 eye-diagrams of the LTX lane for both STD and SiPho modes. The post-processed SNDR/bandwidth are 30.8dB/31GHz and 28dB/35GHz respectively. For the integrated SiPho mode, the driver is connected to the 100Ω oscilloscope load, though optimized for 66Ω. The measured random jitter of the TX path in both STD and SiPho mode is <120fs_{rms}.

The THD, SNR and SNDR of the LRX lane with -1dBFS signal are shown in Fig. 23.2.4 as a function of frequency. SNDR_{nocal} with de-activation of the calibration of sampling time interleaving errors is also reported to replicate the condition of loopback measurements.

Both electrical and optical loopback measurements have been performed with an LTX transmitting to an LRX (Fig. 23.2.5). The DP-16QAM constellations for X and Y polarizations measured in a 1.6T OSFP-XD module running without any optical amplification over 40km of standard single-mode fiber are shown in Fig. 23.2.5. The constellations are recovered at the LRX DSP in an O-band optical setup, with the transmit signal generated using the

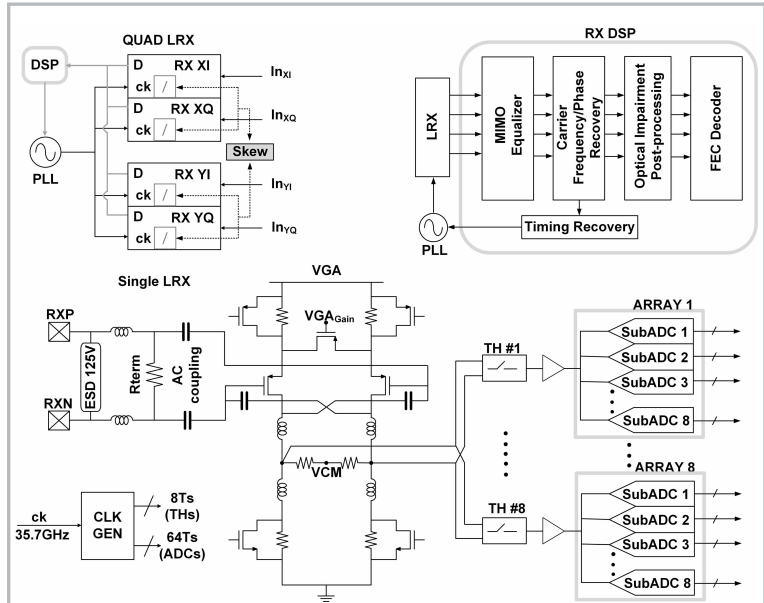


Figure 23.2.3: LRX quad block diagram. Single LRX block diagram. LRX DSP.

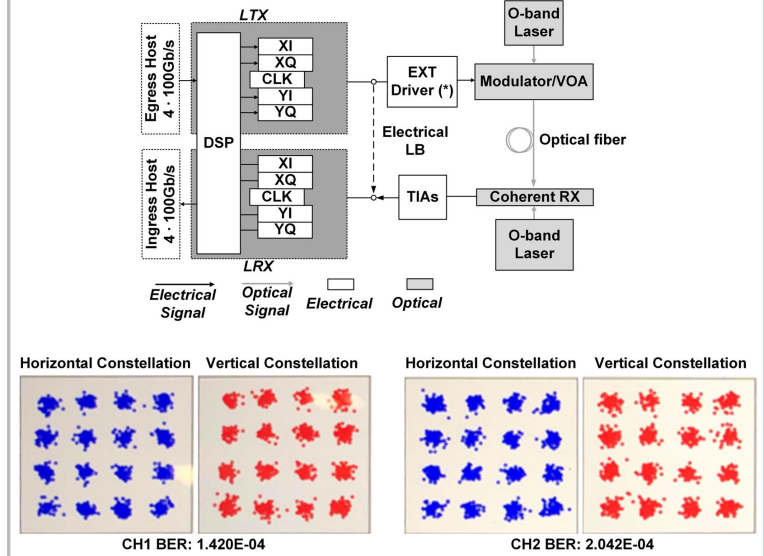


Figure 23.2.5: Loopback setup with (*) EXT driver bypassed in SiPho mode. Measured DP-16QAM constellations.

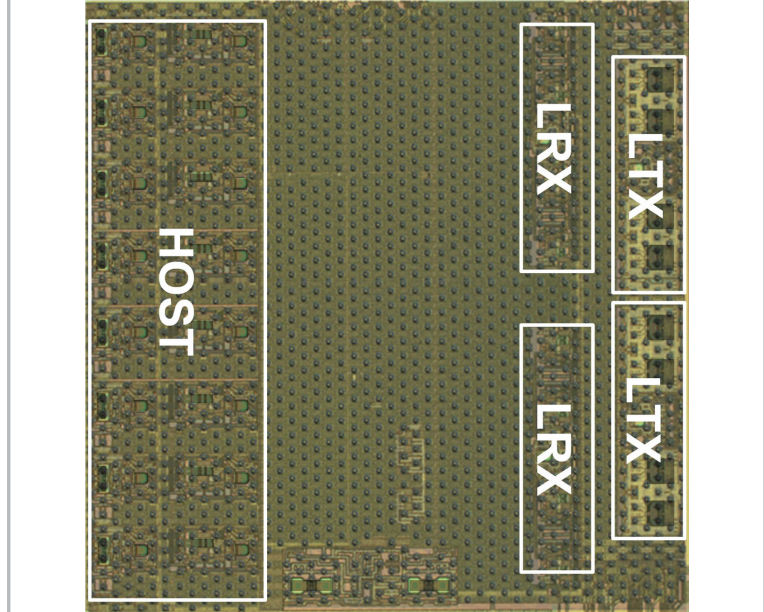


Figure 23.2.7: Die photograph.

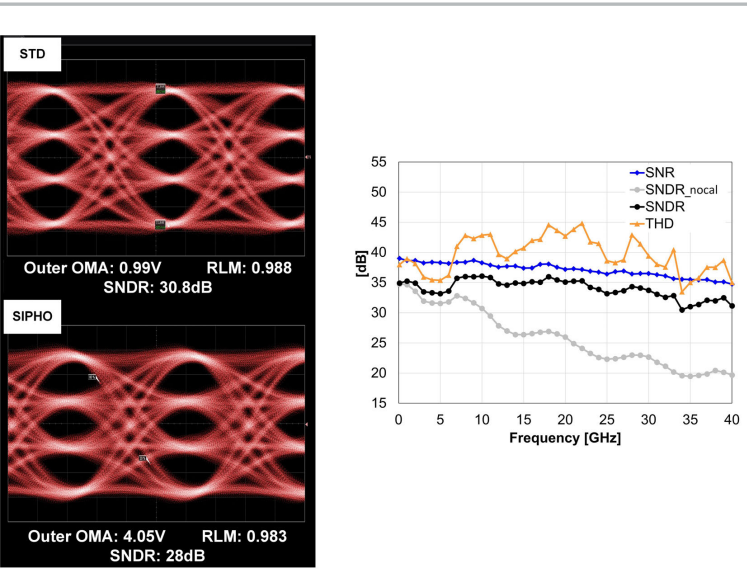


Figure 23.2.4: Measured 125Gb/s LTX eye diagrams (STD and SiPho mode). Measured LTX performance vs input frequency.

	This Work	[8]	[13]	[14]
Channel Datarate [Gb/s]	400	400	600	100
Modulation	DP-16QAM	PS-16QAM	DP-64QAM	DP-QPSK
Process	5n Finfet	7n Finfet	16nm Finfet	20nm
RX and TX resolution [bits]	7	8	8	8
RX Speed [GS/s]	71.4	97	105	64
RX SNDR low freq. [dB]	35	41.1	42.8	39.7
RX SNDR high freq. [dB]	31@35.7G	32@30G	39.2@20.9G	32.5@19G
TX speed [GS/s]	62.5	97	105	64
Optical driver support	SiPho	No	No	No
TX Swing SiPho/STD [Vppd]	4/1	No/1.1	No/1.2	No/0.7
TX Linear Fit SNDR SiPho/STD [dB]	28/30.8	No/NA	No/NA	No/NA
Analog Power SiPho [W]	2.2@400G	No	No	No
Analog Power STD [W]	1.49@400G	3@400G	4.5*@600G	6.3*@100G
Analog Power Efficiency SiPho [pJ/b]	5.5	No	No	No
Analog Power Efficiency STD [pJ/b]	3.72	7.5	7.5*	63*

* TX and RX only

Figure 23.2.6: Comparison table.